

CEVE 421/521 Schedule

i Note

Check the page for each reading to see discussion questions. You will often be encouraged to focus only on part of relevant papers.

Final Project: Deliverables are due throughout the semester (see schedule below).

CEVE 521 students: You will lead one Paper Discussion session. See Graduate Seminar Requirements.

Week	Date	What
1	Mon., Jan. 12	Lecture: Intro to Climate Risk Management []
	Wed., Jan. 14	Topical Discussion: T. Frank [1] and S. Crawford [2]
	Fri., Jan. 16	Lab 1: Julia Setup & Hello World
2	Mon., Jan. 19	<i>Martin Luther King, Jr. Day (No Class)</i>
	Wed., Jan. 21	Lecture: Hazard, Exposure, Vulnerability []
	Fri., Jan. 23	Lab 2: Intro to iCOW
3	Mon., Jan. 26	Lecture: Climate Variability, Extremes, and Change []
	Wed., Jan. 28	Paper Discussion: S. Seneviratne <i>et al.</i> [3]
	Fri., Jan. 30	Lab 3: Sea-Level Rise Scenarios with BRICK / Project: Topic Proposal Due
4	Mon., Feb. 2	Lecture: Uncertainty & Monte Carlo []
	Wed., Feb. 4	Paper Discussion: M. Zarekarizi, V. Srikrishnan, and K. Keller [4]
	Fri., Feb. 6	Lab 4: Monte Carlo Hazard Modeling
5	Mon., Feb. 9	Lecture: Risk Metrics (EAD, VaR, CVaR) []
	Wed., Feb. 11	Lab 5: Probabilistic Risk Analysis
	Fri., Feb. 13	<i>No Class</i>

Week	Date	What
6	Mon., Feb. 16	Exam 1 (Covers Module 1)
	Wed., Feb. 18	Lecture: Valuation & Benefit-Cost Analysis []
	Fri., Feb. 20	Paper Discussion: K. Arrow <i>et al.</i> [5]
7	Mon., Feb. 23	Lecture: Optimal Static Decisions []
	Wed., Feb. 25	Paper Discussion: D. van Dantzig [6]
	Fri., Feb. 27	Lab 6: Simulation-Optimization / Project: Memo 1 Due
8	Mon., Mar. 2	Lecture: Sensitivity & Value of Information []
	Wed., Mar. 4	Paper Discussion: M. C. Runge, S. J. Converse, and J. E. Lyons [7]
	Fri., Mar. 6	Lab 7: Value of Partial Information
9	Mon., Mar. 9	Lecture: Trade-offs & Values []
	Wed., Mar. 11	Paper Discussion: T. M. Daw <i>et al.</i> [8]
	Fri., Mar. 13	Exam 2 (Covers Module 2)
	Mar. 16-20	<i>Spring Break</i>
10	Mon., Mar. 23	Lecture: Robustness & Deep Uncertainty []
	Wed., Mar. 25	Paper Discussion: R. J. Lempert and M. E. Schlesinger [9] and S. H. Schneider [10]
	Fri., Mar. 27	Lab 8: Robustness Metrics / Project: Memo 2 Due
11	Mon., Mar. 30	Lecture: Exploratory Modeling & Scenario Discovery []
	Wed., Apr. 1	Paper Discussion: S. Banks [11]
	Fri., Apr. 3	Lab 9: Scenario Discovery
12	Mon., Apr. 6	Lecture: Sequential Decision Making []
	Wed., Apr. 8	Paper Discussion: G. G. Garner and K. Keller [12]
	Fri., Apr. 10	Lab 10: Real Options & Dynamic Policy Search / Project: Memo 3 Due
13	Mon., Apr. 13	Lecture: Multi-Objective Optimization []

Week	Date	What
	Wed., Apr. 15	Paper Discussion: J. R. Kasprzyk, S. Nataraj, P. M. Reed, and R. J. Lempert [13]
	Fri., Apr. 17	Lab 11: Pareto Fronts / Project: Slides Due
14	Mon., Apr. 20	Final Project Presentations (Teams 1 & 2)
	Wed., Apr. 22	Final Project Presentations (Teams 3 & 4)
	Fri., Apr. 24	Final Project Presentations (Teams 5 & 6)

Bibliography

- [1] T. Frank, "Bold New Jersey Shore Flood Rules Could Be Blueprint for Entire U.S. Coast," *Scientific American*, Aug. 2022, Accessed: Jan. 11, 2023. [Online]. Available: <https://www.scientificamerican.com/article/bold-new-jersey-shore-flood-rules-could-be-blueprint-for-entire-u-s-coast/>
- [2] S. Crawford, "Floods and Storms Are Ravaging the Jersey Shore. Why Do We Keep Building It Back?." Accessed: Dec. 05, 2025. [Online]. Available: <https://www.rollingstone.com/culture/culture-features/floods-storms-jersey-shore-beach-rebuilding-1235477927/>
- [3] S. Seneviratne *et al.*, "Weather and Climate Extreme Events in a Changing Climate," *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, UK and New York, NY, USA, 2021. doi: 10.1017/9781009157896.013.
- [4] M. Zarekarizi, V. Srikrishnan, and K. Keller, "Neglecting Uncertainties Biases House-Elevation Decisions to Manage Riverine Flood Risks," *Nature Communications*, vol. 11, no. 1, p. 5361, Oct. 2020, doi: 10.1038/s41467-020-19188-9.
- [5] K. Arrow *et al.*, "Determining Benefits and Costs for Future Generations," *Science*, vol. 341, no. 6144, pp. 349–350, Jul. 2013, doi: 10.1126/science.1235665.
- [6] D. van Dantzig, "Economic Decision Problems for Flood Prevention," *Econometrica*, vol. 24, no. 3, pp. 276–287, 1956, doi: 10.2307/1911632.
- [7] M. C. Runge, S. J. Converse, and J. E. Lyons, "Which Uncertainty? Using Expert Elicitation and Expected Value of Information to Design an Adaptive Program," *Biological Conservation*, vol. 144, no. 4, pp. 1214–1223, Apr. 2011, doi: 10.1016/j.biocon.2010.12.020.
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- [9] R. J. Lempert and M. E. Schlesinger, "Robust Strategies for Abating Climate Change," *Climatic Change*, vol. 45, no. 3–4, pp. 387–401, Jun. 2000, doi: 10.1023/A:1005698407365.

- [10] S. H. Schneider, "Can We Estimate the Likelihood of Climatic Changes at 2100?," *Climatic Change*, vol. 52, no. 4, pp. 441–451, Mar. 2002, doi: <http://dx.doi.org/10.1023/A:1014276210717>.
- [11] S. Banks, "Exploratory Modeling for Policy Analysis," *Operations Research*, vol. 41, no. 3, pp. 435–449, Jun. 1993, doi: 10/c7rgcr.
- [12] G. G. Garner and K. Keller, "Using Direct Policy Search to Identify Robust Strategies in Adapting to Uncertain Sea-Level Rise and Storm Surge," *Environmental Modelling & Software*, vol. 107, pp. 96–104, Sep. 2018, doi: 10.1016/j.envsoft.2018.05.006.
- [13] J. R. Kasprzyk, S. Nataraj, P. M. Reed, and R. J. Lempert, "Many Objective Robust Decision Making for Complex Environmental Systems Undergoing Change," *Environmental Modelling & Software*, vol. 42, pp. 55–71, Apr. 2013, doi: 10.1016/j.envsoft.2012.12.007.